INCH-POUND

A-A-59321A
October 16, 2003
SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

TEST SET, TRANSPONDER SET

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) describes a portable flight-line tactical air navigation (TACAN) test set (referred to herein as the test set). This CID is meant as a minimum requirement for the test set in which only those manufacturers that meet or surpass the following requirements are supplied per this CID. This test set shall include all necessary accessories needed to test and fault detect the Air Force aircraft TACAN systems referenced herein.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation and be bid sample tested within the accuracies, limits, and specifications herein. Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet this description. The equipment shall be Class 1 in accordance with MIL-PRF-28800, except as specified herein.

2.1 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein except as specified herein.

2.1.1 Water Resistance. The equipment shall not be required to be watertight and splash proof. The equipment shall conform to the drip proof requirements of MIL-PRF-28800, Class 2, while the equipment is not operating and the equipment covers are closed.

2.1.2 Explosive Atmosphere. The equipment shall meet the explosive atmosphere requirements of MIL-STD-810, METHOD 511.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611

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2.1.3 Salt Exposure. Salt exposure verification testing shall not be required.

2.1.4 Operating Temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of -20 degrees C to +50 degrees C.

2.1.5 Relative Humidity. The equipment shall meet its performance and accuracy requirements in an operating environment where the relative humidity is 5 to 75±5 percent in the temperature range of 10 to 30 degrees C and 5 to 45±5 percent in the temperature range of 30 to 50 degrees C.

2.1.6 Altitude Requirements. The equipment shall meet its performance and accuracy requirements operating and non-operating as specified in MIL-PRF-28800, except the maximum required altitude shall be 4000 meters.

2.1.7 Random Vibration. The equipment shall conform to the random vibration requirements of MIL-PRF-28800, Class 2.

2.1.8 Transit Drop. The equipment shall conform to the transit drop requirements of MIL-PRF-28800, Class 2.

2.2 Electrical Power Sources and Connections. The equipment shall operate from nominal commercial, military, and shipboard power sources with a nominal voltage of 120 Vrms single-phase AC (±10%) at line frequencies of 50 Hz, 60 Hz, and 400 Hz (±5%) in accordance with MIL-PRF-28800 paragraph 3.1.5.2 and 3.5.1.2.a. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein. AC input power shall be used to power the signal generator and recharge its internal batteries.

2.3 Internal DC Power. DC power shall be supplied by internal, rechargeable storage batteries. The signal generator shall be capable of operating away from an external AC power source. The rechargeable batteries shall have the capability of operating the unit with full performance for a period of not less than one hour for the operating temperature range specified herein. The batteries shall conform to MIL-PRF-28800.

2.4 Recharge Time. The charger circuitry shall be such that a condition of charge, which shall permit at least one hour of continuous use of the unit, within all specification requirements, shall be reached in no more than eight hours following a low battery indication. Under no condition of discharge shall the charging rate applied to the battery exceed the battery manufacturer's recommended maximum rate, nor shall an indefinite charging period result in any damage to the battery or any other components.

2.5 Battery Conservation. The signal generator shall cycle to the off or standby mode after 15 minutes of switch inactivity to conserve internal battery power. As an alternative, the signal generator shall include a spare battery and charger at no additional cost.

2.6 Battery Charger. A battery charger shall be provided in accordance with MIL-PRF-28800, Section 3.

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2.7 Pre-Conditioning Requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

2.8 Calibration. The calibration interval shall be as specified in MIL-PRF-28800 Section 3. No special tools shall be required for alignment or adjustment except those furnished with the test set.

2.9 Performance Requirements. The instrument shall meet all requirements specified herein, after the warm-up period specified in MIL-PRF-28800, under all combinations of input power conditions, output signal conditions and operating service condition as specified in MIL-PRF-28800.

2.9.1 Controls. The following controls shall be provided: MODE-CHANNEL Select, BATTERY-TEST, PEAK POWER, METER ILLUMINATION, RANGE-BEARING Select, RANGE, NORAMAL, INVERSE.

2.9.2 Indicators. The tester shall provide indicator/indicators for displaying battery status, interrogation and transmission.

2.9.3 Functional Description. The test set shall be capable of testing TACAN systems listed in paragraph 2.12. The test set shall simulate a surface beacon in the normal transmit/receive T/R (X) and (Y) modes of operation and an airborne TACAN in the air-to-air A/A (X) and (Y) modes of operation including A/A range and bearing, and inverse (INV) mode capabilities, for the maintenance of TACAN airborne equipment.

2.9.4 Transmission Methods. The TACAN signals shall be transmitted to and received from the unit under test by means of an internal antenna or by means of a direct coupled coaxial connector.

2.9.5 Operating Channels.

T/R	31 (X)	992 MHz
T/R	91 (X)	1178 MHz
T/R	93 (Y)	1054 MHz
T/R	29 (Y)	1116 MHz
A/A	93 (X,Y)	1054 MHz
A/A	29 (X,Y)	1116 MHz

The output frequency shall be within ±0.002 percent.

2.9.6 RF Output Power. The test set shall have an effective radiated power of -15 dBm and a direct coupled output power of -25 dBm at each frequency specified in paragraph 2.10.5.

2.9.7 OFF, BAT-TEST, Mode Channel Select, Peak Power Measurement. A front panel control shall be provided to initiate operation, test batteries, select T/R(X), T/R(Y), A/A(X), and A/A(Y) modes with fixed channels as specified in para. 2.10.5 and provide a TACAN R/T peak power measurement mode when used with other accessories.

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2.9.8 Range Bearing Selection. A panel control shall provide for eight simulated ranges with eight corresponding fixed bearings. The simulated ranges shall be 0, 5, 68, 125, 194, 283, 297, and 389 nautical miles.

a. Minimum Range – 0.0 nautical miles
b. Maximum Range – 399 nautical miles
c. Ranges selected to nearest nautical mile.

Simulated bearings shall be 0, 45, 90, 135, 180, 225, 270 and 315 degrees.

2.9.9 A/A Range Only, Normal, and Inverse Mode Selection. A control shall be provided for the selection of either A/A Range Only, Normal and Inverse Mode. The Normal Mode shall be used for T/R(X) and T/R(Y) modes of operation, unless Inverse Mode is desired and in the Normal position selection of either A/A(X) or A/A(Y) on the Mode Channel control shall provide both range and bearing capability.

2.9.10 Meter Illumination. A control shall be provided for the illumination of the meter.

2.9.11 RF In/Out. A coaxial connector shall be provided for the purpose of a direct connection to the airborne TACAN under test used in conjunction with available accessories.

2.9.11 North Reference Trigger (NRT)/SYNC Out. A coaxial connector shall be provided for the purpose of supplying a North Reference Trigger during Inverse Mode operation, which shall also be capable of use as a 15 Hz synchronization trigger for testing purposes.

2.9.12 Output Pulses. The output pulse envelope shall be quasi Gaussian pulse whose width shall be 3.5 ±0.5 µ seconds when measured between the points on the leading and trailing edges of the pulse that are 50 percent of the peak amplitude.

2.9.12.1 Main Reference:

T/R(X) - A group of 12 pairs of pulses with the spacing between pairs space 30 ±0.5 µsec.
T/R(Y) - A group of 13 single pulses with the spacing between pairs space 30 ±0.5 µsec.
A/A(X and Y) - A group of 10 single pulses with the spacing between pairs space 30 ±0.5 µsec.

2.9.12.2 Auxiliary Reference:

T/R(X) - A group of 6 pairs of pulses with the spacing between pairs space 24 ±0.5 µsec.
T/R(Y) - A group of 13 single pulses with the spacing between pairs space 15 ±0.5 µsec.

2.9.12.3 Identification Signal: The identification signal is present at all times.

2.9.12.3.1 T/R(X) and T/R(Y) modes: The identification signal shall consist of a series of paired pulses transmitted at a rate of 1350 pulsed pairs per second. An equalizing pulse pair shall be transmitted 100 µsec ±10 µsec after each identification pair.

2.9.12.3.2 A/A(X) and A/A(Y) modes: The identification signal shall consist of a series of single pulses transmitted at a rate of 1350 pulses per second.

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2.9.12.4 Pulse Spacing: The spacing between pulses in a pulse pair for the main reference group, auxiliary reference group, range replies, and identification signal shall be as follows:

T/R(X)	12 ±0.5 µsec
T/R(Y)	30 ±0.5 µsec
A/A(X)	12 ±0.5 µsec
A/A(Y)	24 ±0.5 µsec

2.9.12.5 A/A Range Reply Signal: When in A/A mode, the interrogation shall initiate a single pulse range reply.

2.9.12.6 Zero Mile Delay: The zero mile delay time between the 50% amplitude point of the leading edge of the first interrogation pulse of a coded pair to the 50% amplitude point on the leading edge of the first reply pulse of a coded pair shall be as follows:

T/R(X)	50 ±1.2 µsec
T/R(Y)	56 ±1.2 µsec
A/A(X)	52 ±1.2 µsec
A/A(Y)	74 ±1.2 µsec

2.9.12.7 Range Reply Accuracy: The range reply accuracy from 0 to 399 miles shall be ± 0.1 nautical miles for the selected range positions.

2.9.12.8 Bearing Accuracy: The bearing accuracy shall be ± 1.0 degree for the selected bearing positions in the T/R(X) and T/R(Y) modes and ± 2.0 degrees in the A/A(X) and A/A(Y) modes of operation.

2.9.12.9 Decoding Tolerance: The test shall respond to an interrogation whose coded pairs deviate from nominal by ±1.0 µsec and shall not respond to an interrogation whose coded pulse pairs deviate from nominal by more than ± 4.0 µsec.

2.9.12.10 Air to Air Interrogation: In the A/A mode, the interrogation rate shall be 150 ± 20 pulse pairs per second. In addition, an interrogation shall be provided to indicate when the TACAN airborne unit under test is replying to test set interrogations:

A/A(X)	62 -2 +4 µsec
A/A(Y)	74 -2 +4 µsec

Signal priority is as follows in the order listed:

A. Main Reference Group
B. Auxiliary Reference Group
C. Identification Signal and Range Replies (equal priority)

2.9.12.11 Composite Bearing Signal (15 Hz and 135 Hz): The composite bearing signal amplitude modulates the radio frequency output pulse signals in the T/R(X) and T/R(Y) modes of operation. 15 Hz and 135 Hz sine wave components are as specified in MIL-STD-291, except tolerances shall be as specified herein.

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2.9.12.11.1 15Hz and 135 Hz Modulation: The 15 Hz and 135 Hz modulation shall be 20% ± 10% each and 40% ± 10% composite.

2.10.12.11.2 A/A Mode Bearing Signal: In the A/A(X) and A/A(Y) modes of operation, the 135 Hz component of the composite bearing signal shall be suppressed. The 15 Hz modulation shall be 20% ± 10%.

2.9.12.12 Inverse Mode: The North Reference Trigger shall be made available at a coaxial connector for Inverse A/A(X), Inverse A/A(Y), Inverse T/R(X) and Inverse T/R(Y) modes. The North Reference Trigger shall be capable of shunting a ground voltage of 6 ± 1 volts through an impedance of 1000 Ohms or greater. The 15 Hz bearing shall be determined by the time delay from the rising edge of the North Reference Trigger signal to the negative going zero crossing of the 15 Hz modulating signal. The signal shall be applied for a period of 70 ± 10 µsec at a repetition rate of 15 Hz ±0.2%. The accuracy of the rising edge coinciding with the negative going zero crossing of the 15 Hz modulating signal at zero degrees bearing shall be ± 20 µsec.

2.9.12.13 Operating Capabilities:

2.9.12.13.1 Radiate Mode: The test set shall be capable of checking the operational performance of the airborne TACAN at a distance of 10 to 100 feet from the aircraft antenna. The peak power output of the airborne TACAN can be anywhere from 500 to 4000 watts. The ramp tester shall not suffer damage up to 2 feet from the aircraft antenna.

2.9.12.13.2 Direct Connection (RF IN/OUT): The test set shall be capable of checking the operational performance of the airborne TACAN when connected to its antenna connector. Use an optional attenuator for the protection of input circuitry.

2.9.12.14 Input Power: The maximum peak input power shall be less than +36 dBm peak at the RF In/Out connector.

2.9.12.15 Output Power: The minimum output power shall be -25 ± 3 dBm at the RF In/Out connector.

2.9.12.16 Peak Power and Sensitivity Accessories: Accessories shall be made available for the purpose of peak power and sensitivity measurements with the test set used in its direct connection mode. The accessories shall consist of a Measurement Attenuator and two coaxial cables.

2.9.12.16.1 Measurement Attenuator: The measurement attenuator shall be made available when defined in the procuring document. The device must be used when connecting the tester directly to the antenna of the airborne TACAN. It shall consist of a fixed 30 ± 0.5 dB of attenuation and a variable 49 ± 0.5 dB step attenuator incremented in 1.0 dB steps. With the variable attenuator set to 10 dB, 30 dB of attenuation shall remain and proper operation shall be achieved with a TACAN receiver sensitivity between -65 and -114 dBm and a peak power output of 500 to 4000 watts.

2.9.12.16.1.1 Bearing Sensitivity: The bearing lock-on sensitivity of the airborne TACAN shall be capable of being measured within ± 3 dB at the RF In/Out connector. The indication of

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sensitivity shall be determined by the aircraft bearing indicator changing from search to lock mode with decreasing attenuation.

2.9.12.16.1.2 Peak Power: The peak power output of the airborne TACAN shall be capable of being measured with ± 2 dB at the RF In/Out connector. The indication of peak power shall be determined with the control in the Peak Power measurement position and all power is removed from all other circuitry.

2.9.12.16.2 Accessory Cables: One coaxial cable shall be provided to allow for interconnection from the airborne TACAN to the Measurement Attenuator. Connector types shall be selected as to disallow direct connection between airborne TACAN and the tester. A separate cable provides the North Reference Trigger for inverse mode.

2.10 Physical Characteristics. The equipment shall be portable with the following physical characteristics.

2.10.1 Dimensions. The maximum dimensions for the test set shall allow for portable testing in accordance with Section 6 of MIL-PRF-28800.

2.10.2 Weight. The total weight of the test set, including all accessories (including job guide), antenna, batteries, and battery charger shall not exceed 11 pounds.

2.11 Compatibility. The test set shall be compatible with the following Air Force airborne TACAN equipments: AN/ARN-21(V), AN/ARN-52(V), AN/ARN-62, AN/ARN-65, AN/ARN-72, AN/ARN-84(V), AN/ARN-90, AN/ARN-91, AN/ARN-96, AN/ARN-105, AN/ARN-106, AN/ARN-118, AN/ARN-119, AN/ARN-136, AN/ARN-139, AN/ASQ-19A, AN/ASQ-25, AN/ASQ-37, AN/ASQ-88A, ANS-900A, ANS-905, ANS-915.

2.12 Manuals. The equipment shall be delivered with operator, maintenance, and illustrated parts manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, Section 6, shall be as specified in the contract or order.

3. REGULATORY REQUIREMENTS

3.1 The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and

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all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

4.3 The contracting officer has the option of accepting or rejecting the product.

5. PRESERVATION, PACKING, AND MARKING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Sources of documents.

6.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.2 The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

MILITARY INTEREST

Custodian:

Air Force – 99

Preparing activity:

Air Force - 84

Agent:
Air Force - 99

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